

Learning ML, DL, NLP

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course
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Sec-29 Logistic
Regression

Sec-3 :- Performance
metrics

Sec-4 :- Logistic
Regression OVR

Performance Metrics, Accuracy,
Precision, Recall & F-Score.

① Confusion Matrix

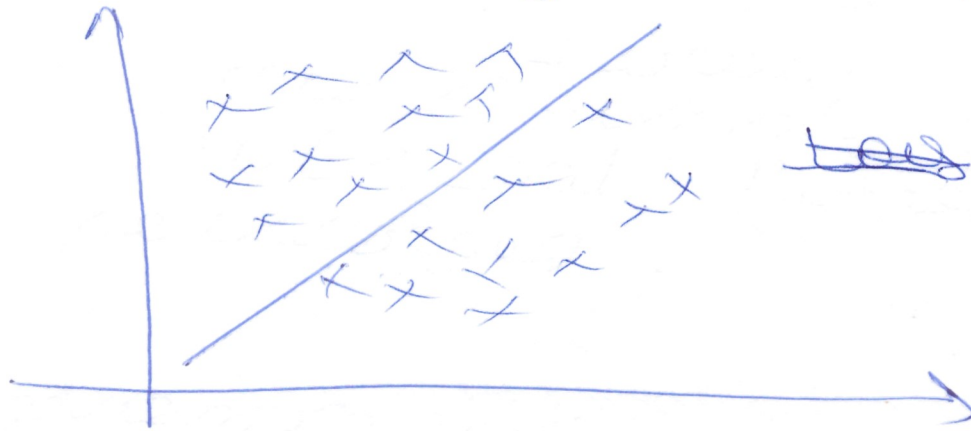
② Accuracy

③ Precision

④ Recall

⑤ F-Score

Logistic Regression



Dataset

x_1	x_2	y o/p	$\hat{y}$ predicted by model
-	-	0	1
-	-	1	1
-	-	0	0
-	-	1	1
-	-	1	1

→ consider this dataset
 with x_1, x_2 as input
 $y, \hat{y}$ as output

① Confusion matrix

in case of Binary classification, this is a 2×2 matrix.

	0	1	Actual values
1			
0			

→ Predicted values.

If both the predicted & actual values are same, we call it correct prediction. If not, we call it wrong prediction.

	True Positive (TP)	False Positive (FP)
1	True Positive (TP)	False Positive (FP)
0	False Negative (FN)	True Negative (TN)

Accuracy :-

Hence the formula for accuracy can be stated as:

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN}$$

$$= \frac{3 + 1}{3 + 2 + 1 + 1}$$
$$= \frac{4}{7}$$

(A) Dataset

→ 1000 data points
(Binary Classification)

90% Accuracy

900

100

0

Imbalanced dataset

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Precision

→ ~~if we have~~ ~~an~~ ~~important~~ ~~dataset~~,
then ~~it's~~ ~~better~~ ~~to~~ ~~not~~
use ~~it~~ ~~as~~ ~~a~~ ~~guess~~ ~~is~~
very ~~high~~ ~~accuracy~~

$$\text{Precision} = \frac{TP}{TP + FP}$$

	1	0	Actual
1	TP	FP	
0	FN	TN	

Out of all
the actual
values, how
many are
correctly
predicted.

④ Recall = $\frac{TP}{TP + FN}$

Out of all
predicted values, how
many are correctly predicted.

use case

Spam Classification

mail → spam

model → spam

This is a true
positive case
as the model
~~was~~ predicted correctly

mail → not spam

model → spam

Blunder

Spam Classifier

mail → Not spam

model → spam

Blunder

$$\text{precision} = \frac{TP}{TP + FP}$$

is it right whether the
specification is correct
or not.

truth $\rightarrow$ diabetes
model $\rightarrow$ not
diabetes

blunder

truth $\rightarrow$ diabetes
model $\rightarrow$ diabetes

good

truth $\rightarrow$ not really
model $\rightarrow$ diabetes

④ tomorrow the stock
market will crash
or not.

Reducing $\rightarrow$ P $\downarrow$ or F $\downarrow$

$$F\text{-}Beta\text{ score} = (1 + \beta^2) \times$$

$$\frac{Precision \times Recall}{Precision + Recall}$$

2) $F\beta$ & FN are both important

$$\beta = 1$$

$$F1\text{ score} = (1 + 1^2) \frac{Precision \times Recall}{Precision + Recall}$$

$$= 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

2) $F\beta$ is more important than FN

$$\beta = 0.5$$

$$F(0.5\text{ score}) = (1 + 0.25) \frac{P \times R}{P + R}$$

2) $FN \gg FP$

$$\beta = 2$$

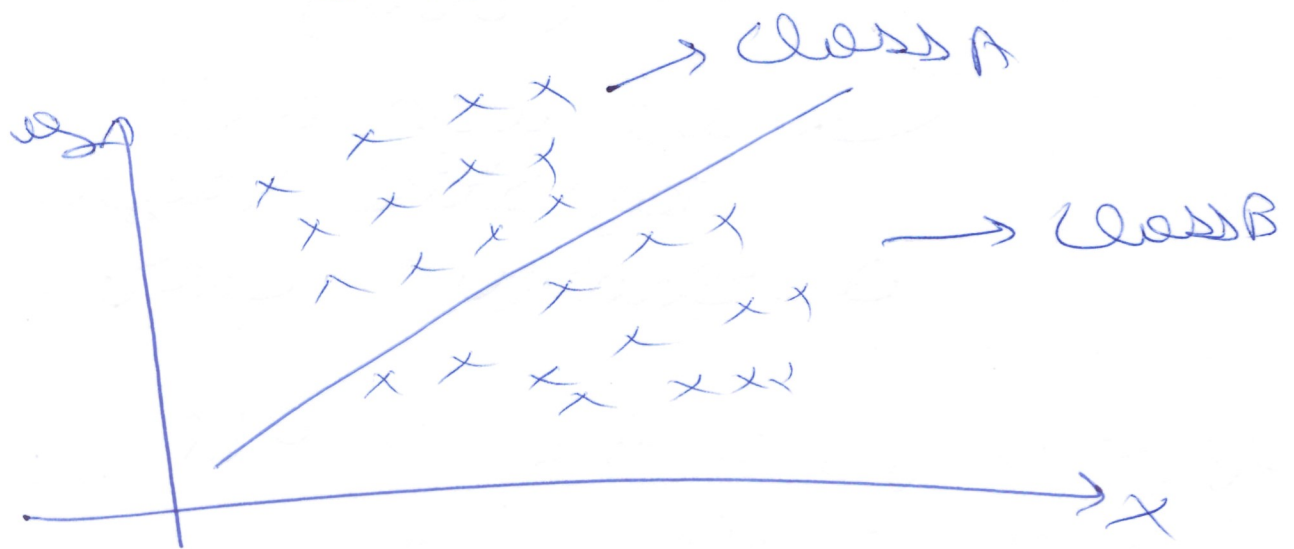
$$\Rightarrow S\left(\frac{P \times R}{P + R}\right) = (1 + 4) \frac{P \times R}{P + R}$$

Avg

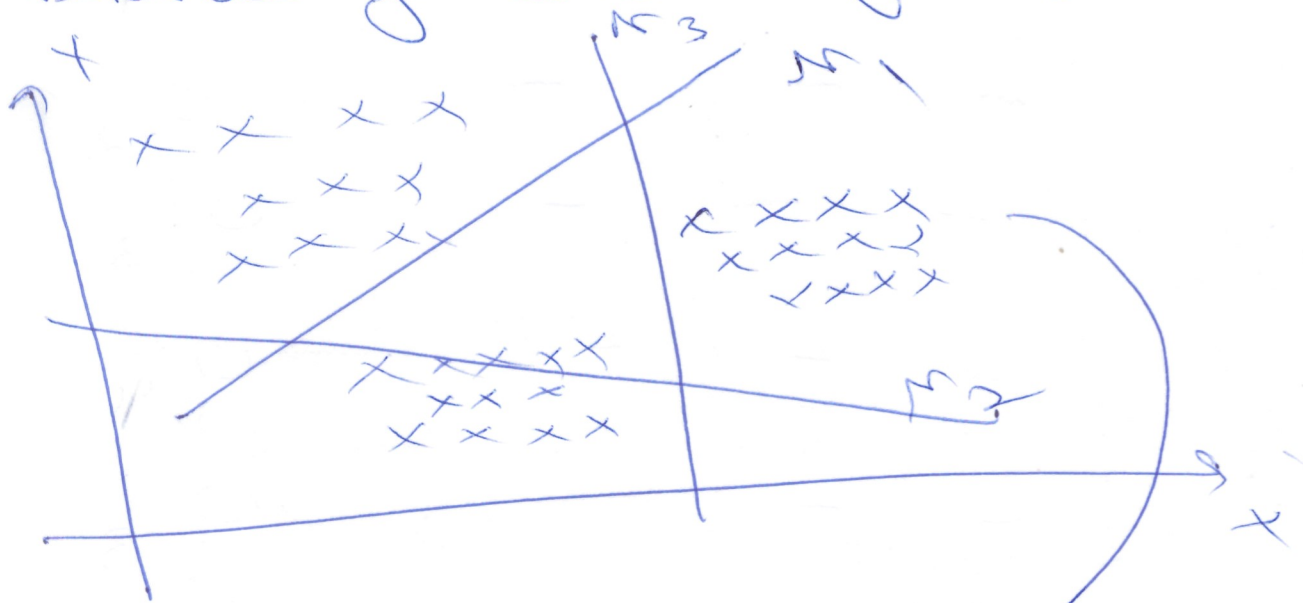
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Logistic Regression (OVR)

One versus Rest



① Binary Classification



Multi Class
Classification
(Logistic Regression)

One important parameter
in Logistic Regression is
One vs Rest (OVR)

→ we use the
~~Logistic~~ Regression
technique here.

b_1	b_2	b_3	O/P
—	—	—	0, 1
—	—	—	0, 3
—	—	—	0, 3
—	—	—	0, 4
—	—	—	0, 3
—	—	—	0, 2

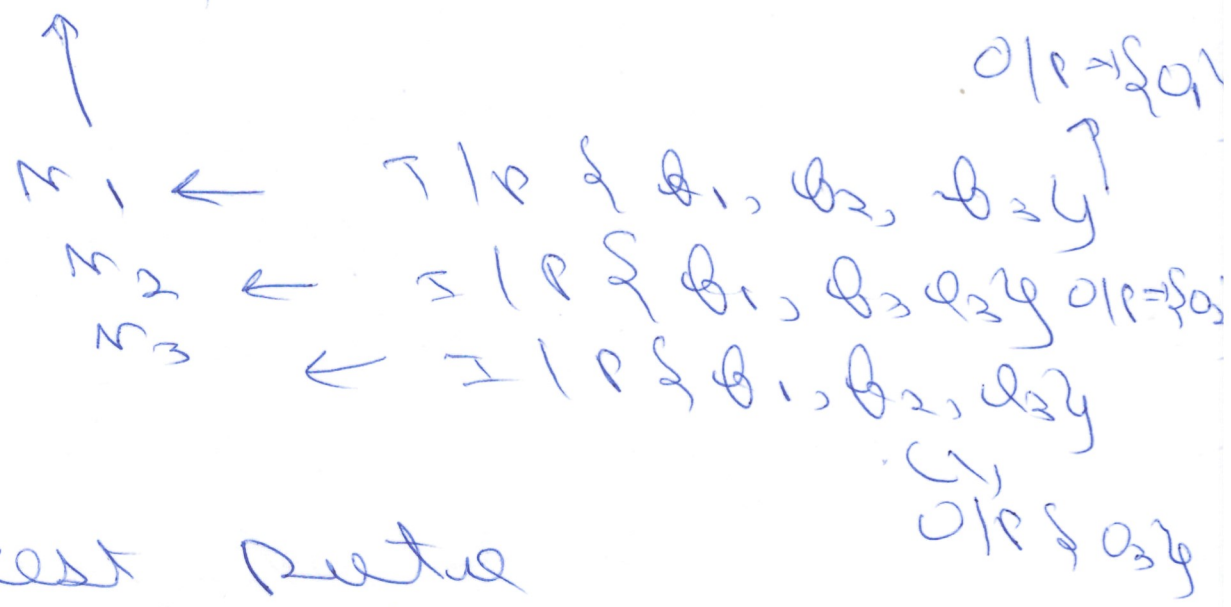
→ M_1 → Binary Classification

M_2 → Binary Classification

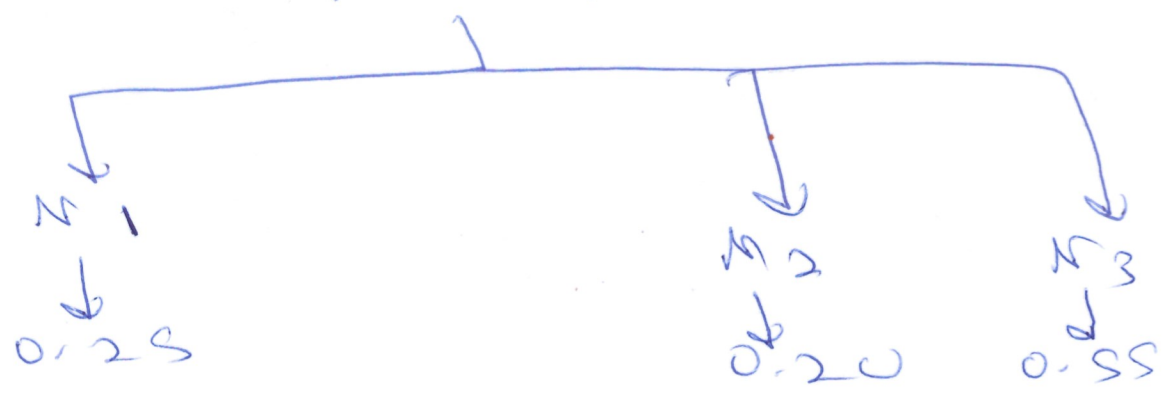
M_3 → Binary Classification

ϕ_1	ϕ_2	ϕ_3	O/P	ϕ_1	ϕ_2	ϕ_3
-	-	-	ϕ_1	1	0	0
-	-	-	ϕ_2	0	1	0
-	-	-	ϕ_3	0	0	1
-	-	-	ϕ_1	1	0	0
-	-	-	ϕ_3	0	0	1
-	-	-	ϕ_2	0	1	0

Model



New test pattern



[0.29, 0.20, 0.51]

↑ ↑ ↑

μ_1 μ_2 μ_3

Highest Probability

2nd taken as output.